Maame Yaa Twumasi

Richmond, VA, open to relocation | maameyaamtwumasi@gmail.com | linkedin.com/in/mytwumasi | github.com/Maame-Yaa | maameyaa.vercel.app

Full Stack Engineer shipping production Next.js and TypeScript forms with server-side validation and rate limiting. Built Python and FastAPI backends and LLM-integrated features for a mortgage platform, including a 7-entity PostgreSQL schema, an 11-state application lifecycle, and the explanation layer of a multi-layer AI agent system. AWS Certified Cloud Practitioner.

TECHNICAL SKILLS

Languages and Frameworks: Python, JavaScript, TypeScript, React, Next.js, Vue.js, Nuxt.js, Node.js, Express, FastAPI, Socket.IO, TensorFlow
Backend and Data: PostgreSQL, MySQL, REST APIs, Zod, SQLAlchemy ORM, Pydantic, Prisma, Database schema design
Cloud and DevOps: Amazon Web Services (AWS), Docker, Git, GitHub/GitLab workflows, Google Apps Script, Automated testing (Pytest), Alembic migrations
Frontend and UI: Tailwind CSS, Figma (design-to-code), NextAuth.js, Stripe

CERTIFICATIONS

Amazon Web Services: AWS Certified Cloud Practitioner

WORK EXPERIENCE

Technical Operations Specialist | Collect Virginia LLC (web development for a growth agency) | Contract *07/2026 - Present*

- Replaced phone and in-person grant intake for nonprofits and small businesses with an 11-section, 53-field form in Next.js, TypeScript and Tailwind CSS, live since August 2026, with conditional sections and Zod validation.
- Built the backend submission API, re-validating every field server-side with a 1MB request cap and IP rate limiting of 10 requests per 10 minutes.
- Routed each submission through a consent check and sanitization to an SMTP email and a 53-column Google Sheets record through Apps Script.
- Added reCAPTCHA v2 to both production forms within a week of a bot attack, with one shared server-side helper verifying each token before any email or record is created. No spam reported since.
- Built a case studies section on the agency's Next.js 16 and TypeScript site, with a listing page, a reusable detail template and three client case studies.

Software Engineer | Cher Inc (mortgage fintech startup) | Contract, Remote *10/2025 - 04/2026*

- Contributed to a multi-layer AI agent system for document requests. Designed the explanation layer, priority ordering, tone control, and LLM context formatting that generates user-facing upload instructions based on application progress.
- Designed a 7-entity PostgreSQL schema in SQLAlchemy for an in-development CRM and Loan Origination System, with soft deletes, indexing and Alembic migrations.
- Built an 11-state application lifecycle as a four-layer state machine, enforcing 13 valid transitions with automatic status updates and progress tracking from document completion.
- Built a document upload API with FastAPI and AWS S3, with 10MB limits, MIME type validation, UUID-based filenames and automatic state transitions. Wrote 30 Pytest tests for upload and state flows.
- Built the frontend authentication foundation in Next.js with NextAuth.js and Prisma, including login, registration, protected dashboard routes, and role-based session handling against the shared PostgreSQL database.
- Surfaced 10 critical issues, including unauthenticated endpoints, by expanding the backend security audit to 35 items with a teammate, and wrote the upload endpoint auth fixes.
- Ran backend services in Docker Compose and built FastAPI endpoints with Pydantic validation, keeping business logic in a service layer apart from API routes.

Software Engineer | Aortem | Contract, Remote *03/2025 - 01/2026*

- Developed frontend features in sprints with daily team meetings for a Vue 3 and Nuxt 3 SaaS admin platform spanning 4 product modules across Standard, Enhanced, and Enterprise subscription tiers.
- Built 10+ reusable Vue components and shared composables for dashboards, tables, filters, pagination, and multi-step modal workflows from Figma designs.
- Implemented Stripe checkout and subscription onboarding flows with conditional routing and gated UI behavior based on user role and plan type.
- Blocked deep links to gated pages with route middleware, backed by one shared composable for plan capability flags and plan-based Stripe portal routing.

Website Developer | The Dayton Weekly News | Part-time, Remote *03/2023 - 04/2025*

- Rebuilt the organization's website, implemented SEO improvements including metadata restructuring, and used Google Analytics to inform layout updates, contributing to a 78% increase in homepage traffic. Google search became the top traffic source.

Software Developer | Camara Web Development | Remote, Founder, Part-time *06/2021 - Present*

- Co-founded a web development business serving small businesses in Ghana. Built websites with MySQL, Express, React, Node.js, WordPress and WooCommerce.
- Delivered a full site build and migration for an electrical supply company, including manual database migration, hosting diagnostics, WooCommerce product catalog setup with bulk image processing, and ongoing client communication. Integrated the DPO Pay payment gateway.

PROJECTS

Elopement Tracker for Children with Autism (MySQL, Express, React, Node.js, IoT, Google Cloud, MQTT, LoRaWAN)

- Built a real-time geofencing alert system with MySQL, Express, React, Node.js, GPS, and LoRaWAN to track children with autism.
- Used React and Google Maps API via Google Cloud for live fence drawing, and Socket.IO and MQTT via The Things Stack for instant alerts.

Bilingual FAQ Chatbot (Python, Streamlit, AI, TensorFlow)

- Created a bilingual chatbot with Python, NLP, and langdetect to support English and French queries. Trained a TensorFlow/Keras intent classification model and delivered a responsive Streamlit UI. Used NLTK tokenization and lemmatization with custom JSON intents for admissions FAQs, reaching 75% accuracy.

EDUCATION

B.Sc. Computer Engineering, Academic City University, Ghana